Riley's sample-size criteria

and why one model per artery (LAD, CX, RCA) is not feasible

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The short answer

- Riley's criteria give the **minimum total sample size** n for a prediction model. The **event fraction** and the **expected strength of the signal** are inputs, so the answer can also be expressed as *events per parameter*, but it is not a fixed "10 events per variable" rule.
- Turned around, they give the **maximum number of parameters** a data set can support.
- Our model needs about **17 parameters** per artery (11 for risk factors, calcium and follow-up time, plus 6 for geometry). At a realistic C-statistic of 0.70 the LAD supports about **4**, the CX and RCA about **10** each. Pooling the three arteries supports about **16**, close to the 19 the pooled model needs.

1 What problem the criteria solve

A prediction model fitted to too few patients *overfits*: it learns noise in the development data. The predicted risks become too extreme (high risks too high, low risks too low), performance in new patients is worse than it looked, and tests of added value (here: geometry) lose their meaning. The old rule of thumb was "at least 10 events per candidate parameter" (EPV ≥ 10). Riley et al. (Stat Med 2019; BMJ 2020) replaced it with three explicit criteria. Their point is that the data needed depends on more than the number of events. It also depends on how hard the outcome is to predict.

Parameters, not variables. P counts *degrees of freedom*, not variables: a continuous variable with a 3-knot spline costs 2, a 3-level factor costs 2, an interaction costs 1 or more. Variables you tried and dropped still count.

2 The three criteria for a binary outcome

Notation: n = number of units (here: plaque-free trunks), ϕ = event fraction (share developing plaque), P = number of parameters, R^2_{CS} = anticipated Cox–Snell R^2 of the model.

1. **Little overfitting.** The expected shrinkage factor must be at least $S = 0.9$ (predictions need shrinking by at most 10%):

$$n \;\geq\; \frac{P}{(S-1)\,\ln\!\left(1 - R^2_{CS}/S\right)}.$$

2. **Small optimism in R^2.** The apparent and optimism-adjusted Nagelkerke R^2 may differ by at most $\delta = 0.05$. Same formula, with S chosen from δ and the maximum possible R^2_{CS}.

3. **Precise overall risk.** The average risk must be estimated within ± 0.05:

$$n \;\geq\; \left(\frac{1.96}{0.05}\right)^2 \phi\,(1-\phi).$$

The required n is the largest of the three. In every scenario in this note, criterion 1 decides the number of parameters.

So is it about sample size or events?

Both, through two routes:

- The largest R^2_{CS} any model can reach is $\max R^2_{CS} = 1 - \left(\phi^{\phi}(1-\phi)^{1-\phi}\right)^2$. A rare outcome has a low ceiling, so the anticipated R^2_{CS} is small and criterion 1 demands a large n. That is why, for rare outcomes, the result looks like an events rule.
- Criterion 3 uses ϕ directly.

In practice the *rarer class* is what limits you. With a 5% event rate that is the events. With our roughly 40% event rate, events and non-events are nearly balanced, so the total number of trunks matters almost as much as the number of events.

3 From an expected C-statistic to R^2_{CS}

Nobody reports R^2_{CS} for plaque outcomes, but C-statistics (AUCs) are reported. Riley, Van Calster and Collins (2021) give a way to convert an anticipated C into R^2_{CS} by assuming the linear predictor is normally distributed in events and non-events. I used that conversion with $\phi \approx 0.39$:

Anticipated C-statistic	0.65	0.70	0.75
R^2_{CS}	0.066	0.116	0.178
Max parameters per 100 trunks (criterion 1)	0.76	1.37	2.20
Trunks needed per parameter	132	73	46

Which C is realistic? PARADIGM's clinical-only model for plaque progression reached 0.684, and the PARADIGM risk score dropped from 0.73 to 0.67 on validation. In CGPS, baseline calcium alone gives a vessel-level AUC of about 0.70. A full model at **0.70 to 0.75** is plausible; 0.70 is the cautious planning value. A weaker signal means fewer parameters for the same data: the same trunks buy almost three times as many parameters at C = 0.75 as at C = 0.65.

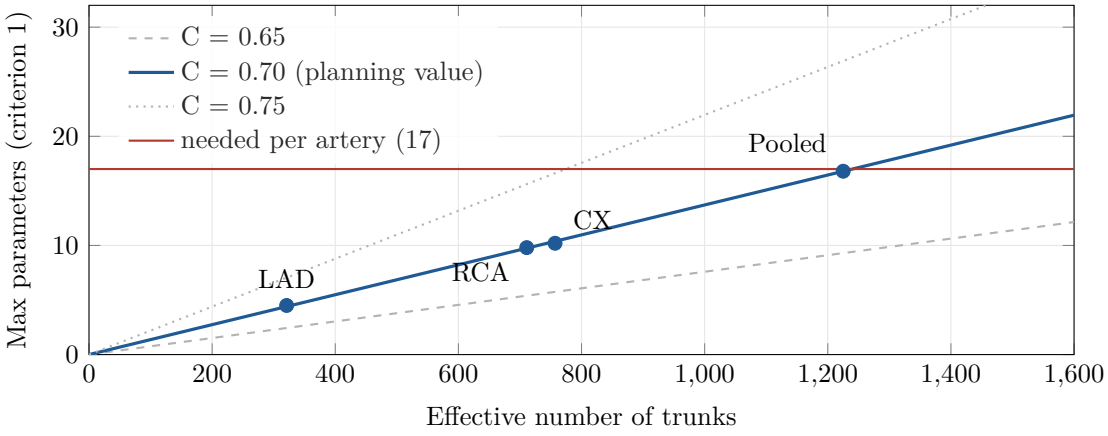
4 What our model has to pay for

Block	Per-artery model	Pooled model
SCORE2 risk factors (age spline, sex, smoking, SBP, cholesterol, diabetes)	7	7
log(1 + total CAC), plaque elsewhere	2	2
log(follow-up years)	1	1
`rand_group` 2 indicator	1	1
Artery (LAD / CX / RCA)	–	2
Base model	**11**	**13**
Geometry block (the thesis question)	6	6
Full model	**17**	**19**

5 The numbers per artery

Inputs come from the aggregate `report.md`, section "plaque-free → plaque" at vessel level: vessels plaque-free at baseline and how many developed plaque by follow-up. Each person has at most one LAD, one CX and one RCA, so a per-artery model has no clustering. For the pooled model, trunks from the same person are correlated (ICC 0.272, design effect 1.46), so the 1 789 vessels count as about $1\,789/1.46 \approx 1\,225$ independent ones.

Model	Trunks	Events	ϕ	Effective n	Max parameters (criterion 1) C = 0.65	C = 0.70	C = 0.75
LAD only	321	139	0.43	321	2	**4**	7
CX only	757	277	0.37	757	5	**10**	16
RCA only	711	281	0.40	711	5	**9**	15
Pooled LAD+CX+RCA	1 789	697	0.39	1 225	9	**16**	26
Needed					17 per artery, 19 pooled		



Worked example: the LAD

321 plaque-free LADs, 139 developed plaque ($\phi = 0.43$). At C = 0.70, $R^2_{CS} = 0.119$, so criterion 1 gives

$$P_{\max} = n\,(S-1)\ln\!\left(1 - \frac{R^2_{CS}}{S}\right) = 321 \times (-0.1) \times \ln(1 - 0.132) \approx 4.5.$$

That is **4 parameters**, against 11 for the base model alone and 17 with geometry. In events per parameter that is 139/4 ≈ 35, far stricter than the old "10 EPV" rule (which would have allowed 13), because the signal is weak.

The LAD also fails criterion 3: estimating the average risk to ±0.05 needs at least 377 trunks, and there are 321.

How many trunks would one artery need?

Solving criterion 1 for n with $P = 17$ and C = 0.70:

	Needed trunks	Needed events	Available trunks	Shortfall
LAD	≈ 1 200	≈ 520	321	×3.7
CX	≈ 1 270	≈ 460	757	×1.7
RCA	≈ 1 230	≈ 490	711	×1.7

What goes wrong if you fit them anyway

- Predicted risks are too extreme and need far more than 10% shrinkage (miscalibration).
- The apparent C-statistic is optimistic and will not hold in new patients.
- A test of the 6-df geometry block is badly underpowered, so "no added value" means nothing, and a positive result is likely to be noise.
- Three separate geometry tests need a multiple-testing correction, which costs power again.

6 Why pooling helps, and its limits

One model on all three arteries spends the base-model parameters *once* instead of three times. Risk factors, calcium and follow-up time act on the person, so their effects are estimated from everyone. The arteries can still differ:

- an **artery main effect** (2 df) gives each artery its own baseline risk;
- **geometry standardised within artery** puts "high curvature" on one scale for LAD, CX and RCA, so one shared slope per feature is meaningful;
- **1 to 3 prespecified artery × geometry interactions** test whether an effect really differs between arteries;
- **per-artery estimates** can be shown descriptively, with wide confidence intervals.

Even pooled, the budget is tight: 16 parameters at C = 0.70 against 19 needed. It fits at C ≈ 0.72 or above, which is plausible given that calcium alone reaches 0.70. Otherwise, trim the model, for example by reducing the geometry block to 3 to 4 features, or use penalisation (ridge).

7 Caveats

- **Proxy numbers.** The counts are vessel-level from `report.md`: they include side branches and do not apply the left-main rule. The chosen trunk design (`artery_design()`, section 0 of `cgps_stats.py`) was added after the report was generated. Re-run it to get the real counts; the LAD will most likely shrink further.
- **The unread half.** Only about 1 020 of 2 116 participants have baseline segment reads. If the other 1 094 turn out to be genuinely plaque-free at baseline, every artery's budget roughly doubles or more. CX and RCA models might then become possible; the LAD would still be tight.
- **Clustering.** Dividing by the design effect is a heuristic, not part of Riley's published method; state it as an assumption.
- **Verify.** Use the `pmsampsize` package (R or Stata) with the final event fraction and an anticipated C, and report the inputs in the thesis.

References

- Riley RD et al. Minimum sample size for developing a multivariable prediction model: Part II – binary and time-to-event outcomes. *Stat Med* 2019;38:1276–96.
- Riley RD et al. Calculating the sample size required for developing a clinical prediction model. *BMJ* 2020;368:m441.
- Riley RD, Van Calster B, Collins GS. A note on estimating the Cox–Snell R^2 from a reported C statistic (AUROC) to inform sample size calculations for developing a prediction model with a binary outcome. *Stat Med* 2021;40:859–64.
- Lee SE et al. (PARADIGM) Clinical, imaging and laboratory predictors of plaque progression, 2020 (clinical-only C = 0.684).